9 A particle moves along the curve $y = \frac{3(x+6)}{x+4}$, where $x \neq -4$, in such a way that the y-coordinate of the particle is increasing at a constant rate of $\frac{4}{27}$ units per second.

$\frac{dy}{dt} = \frac{4}{27}$ units/s

Find the x-coordinates of the particle at that instant that the x-coordinates of the particle is decreasing at a rate of 2 units per second. [5]

$\frac{dx}{dt} = -2$ units/s

$y = \frac{3x+18}{x+4}$

Let $u = 3x+18$, $v = x+4$

$\frac{du}{dx} = 3$ $\quad$ $\frac{dv}{dx} = 1$

$\therefore \frac{dy}{dx} = \frac{3(x+4) - (3x+18)}{(x+4)^2}$

$= \frac{3x+12-3x-18}{(x+4)^2}$

$= \frac{-6}{(x+4)^2}$ M1

$\frac{dy}{dt} = \frac{dy}{dx} \times \frac{dx}{dt}$ M1

$\frac{4}{27} = \frac{-6}{(x+4)^2} \times (-2)$

$\frac{12}{(x+4)^2} = \frac{4}{27}$

$4(x+4)^2 = \cancel{108}\ 324$ M1

$(x+4)^2 = 81$

$x+4 = 9$ or $x+4 = -9$

$x = 5$ $\quad$ $x = -13$ A2

[Turn over